

52 An Introduction to Ecology and the Biosphere

KEY CONCEPTS

- 52.1** Earth's climate varies by latitude and season and is changing rapidly p. 1167
- 52.2** The distribution of terrestrial biomes is controlled by climate and disturbance p. 1171
- 52.3** Aquatic biomes are diverse and dynamic systems that cover most of Earth p. 1177
- 52.4** Interactions between organisms and the environment limit the distribution of species p. 1178
- 52.5** Ecological change and evolution affect one another over long and short periods of time p. 1187

Study Tip

Make a table: As you read the chapter, build a table listing factors that influence species distribution in terrestrial and aquatic environments. Add figure numbers or page numbers that give examples of these factors in each type of environment.

Type of environment	Factor affecting what species live there	Example
Lake	Amount of sunlight that reaches organisms	Photosynthetic plants and algae live at shallow edges and sunlit surface (Figs. 52.14 and 52.16)

➔ Go to Mastering Biology

For Students (in eText and Study Area)

- Get Ready for Chapter 52
- Figure 52.4 Walkthrough: Seasonal Variation in Sunlight Intensity

For Instructors to Assign (in Item Library)

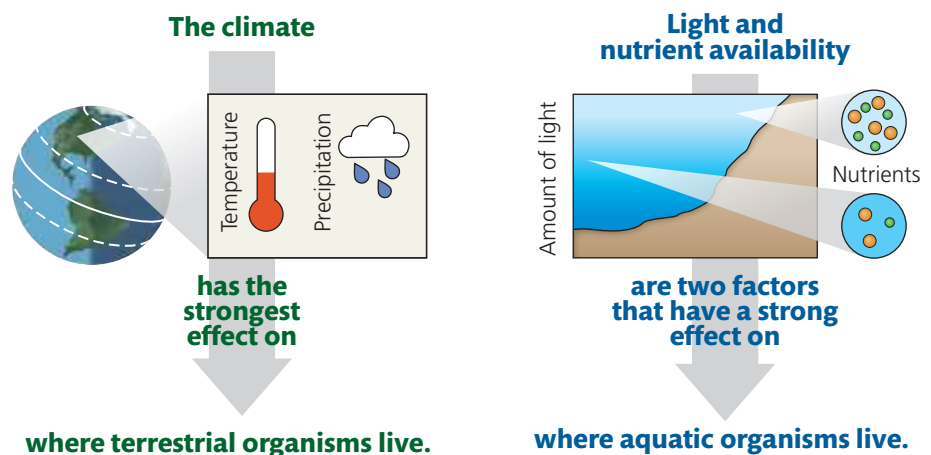
- Interpreting Data: Cholera and Sea Surface Temperature
- Scientific Thinking: Carbon Dioxide and the Growth of Forest Stands



Figure 52.1 Dwarfed by a dime, this tiny frog (*Paedophryne swiftorum*) was discovered on a 2008 expedition to Papua New Guinea. The entire genus *Paedophryne* is known only from a single peninsula in the eastern part of the country. Adult frogs in this genus are about 8 mm long and are among the smallest adult vertebrates on Earth.



What determines where a species such as this tiny frog lives?



Dispersal and interactions among organisms, such as competition, also affect where species are found.